

A Survey on the Imbalanced Data Problem in Medical Diagnosis

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Abstract

Machine learning has become a cornerstone of modern medical diagnosis systems, enabling automated disease detection and clinical decision support. However, medical datasets are frequently characterized by severe class imbalance, where instances of healthy patients vastly outnumber those with a target disease. This imbalance critically undermines the performance of standard classifiers, leading to biased models with unacceptably high false-negative rates—a perilous outcome in healthcare.

This survey provides a comprehensive review of the imbalanced data problem within the context of medical diagnosis. It analyzes the root causes and detrimental consequences of class skew and argues for the necessity of moving beyond accuracy-based evaluation. The paper systematically examines two principal solution strands: data-level methods, including resampling techniques like SMOTE, and algorithm-level approaches, such as cost-sensitive learning and ensemble methods. Real-world applications in cancer detection and chronic disease prediction are reviewed to illustrate the practical efficacy of these strategies.

The synthesis of the literature indicates that hybrid methodologies, which intelligently combine resampling with advanced algorithmic adjustments, yield the most robust and reliable diagnostic performance. Effectively addressing data imbalance is not merely a technical refinement but a fundamental prerequisite for developing trustworthy, equitable, and clinically actionable AI-driven healthcare systems.

1. Introduction

A. Background on Machine Learning in Medical Diagnosis

The integration of Machine Learning (ML) into healthcare has revolutionized diagnostic processes, offering tools for early detection, risk stratification, and personalized treatment planning. Supervised learning paradigms, in particular, are extensively applied to labeled clinical data for tasks ranging from medical image analysis to predictive analytics for conditions like diabetes and heart disease [1].

A pervasive and critical challenge in this domain is the inherent class imbalance of medical datasets. These datasets often reflect real-world disease prevalence, where the number of confirmed positive cases (e.g., a specific cancer) is orders of magnitude smaller than negative or healthy cases. This skew presents a fundamental obstacle, as most standard ML algorithms are designed under the assumption of balanced class distributions, causing them to be biased toward the majority class [2].

B. Problem Significance: Imbalanced Data in Medical Diagnosis

In medical diagnostics, the minority class is almost invariably the class of primary clinical interest (e.g., diseased patients). Misclassifying a positive case as negative (a false negative) can lead to delayed intervention, disease progression, and potentially fatal outcomes. Conversely, while false positives are undesirable, they typically trigger further investigation rather than a missed diagnosis [3].

Models optimized for overall accuracy on imbalanced data can achieve misleadingly high scores (e.g., 95%) by simply predicting the majority class for all instances, thereby failing entirely to identify the critically important minority class. This renders the problem of imbalanced data not just a statistical nuisance, but a profound ethical and clinical concern with direct implications for patient safety [4].

C. Research Objectives

The primary objectives of this survey are threefold:

1. To formally define and characterize the imbalanced data problem within the specific constraints of medical diagnostic applications.
2. To analyze the multifaceted challenges and performance distortions introduced by class imbalance on conventional and modern ML models.
3. To provide a structured review and comparative analysis of the predominant solution strategies proposed in the literature, categorizing them into data-level and algorithm-level approaches.

This work aims to serve as a consolidated reference for students, researchers, and practitioners developing ML models for equitable and effective medical diagnosis.

2. Methodology

This survey was conducted following a systematic literature review methodology to ensure comprehensive and unbiased coverage of the topic. The process is summarized in Figure 1.

A. Literature Search Strategy

A targeted search was executed across major academic databases and digital libraries:

- * IEEE Xplore
- * SpringerLink
- * ScienceDirect (Elsevier)
- * PubMed/MEDLINE
- * ACM Digital Library
- * Google Scholar (for snowballing and identifying seminal works)

The search utilized a combination of keywords and Boolean operators: ("imbalanced data" OR "class imbalance") AND ("medical diagnosis" OR "healthcare" OR "clinical") AND ("machine learning" OR "classification"). The search was restricted to English-language, peer-reviewed articles published between 2000 and 2025.

B. Inclusion and Exclusion Criteria

To maintain relevance and quality, studies were filtered based on the following criteria:

* Inclusion Criteria:

- * Studies explicitly addressing class imbalance in a medical/healthcare diagnostic context.
- * Papers proposing or evaluating a technical solution (data-level, algorithm-level, or hybrid).
- * Foundational papers introducing key algorithms (e.g., SMOTE, cost-sensitive learning).
- * Recent survey or review papers on imbalanced learning or ML in medicine.

* Exclusion Criteria:

- * Studies not involving classification tasks (e.g., regression, clustering only).
- * Applications outside healthcare (e.g., fraud detection, maintenance).
- * Articles without empirical validation or clear methodological description.
- * Non-peer-reviewed publications (preprints, theses, unless highly cited and seminal).

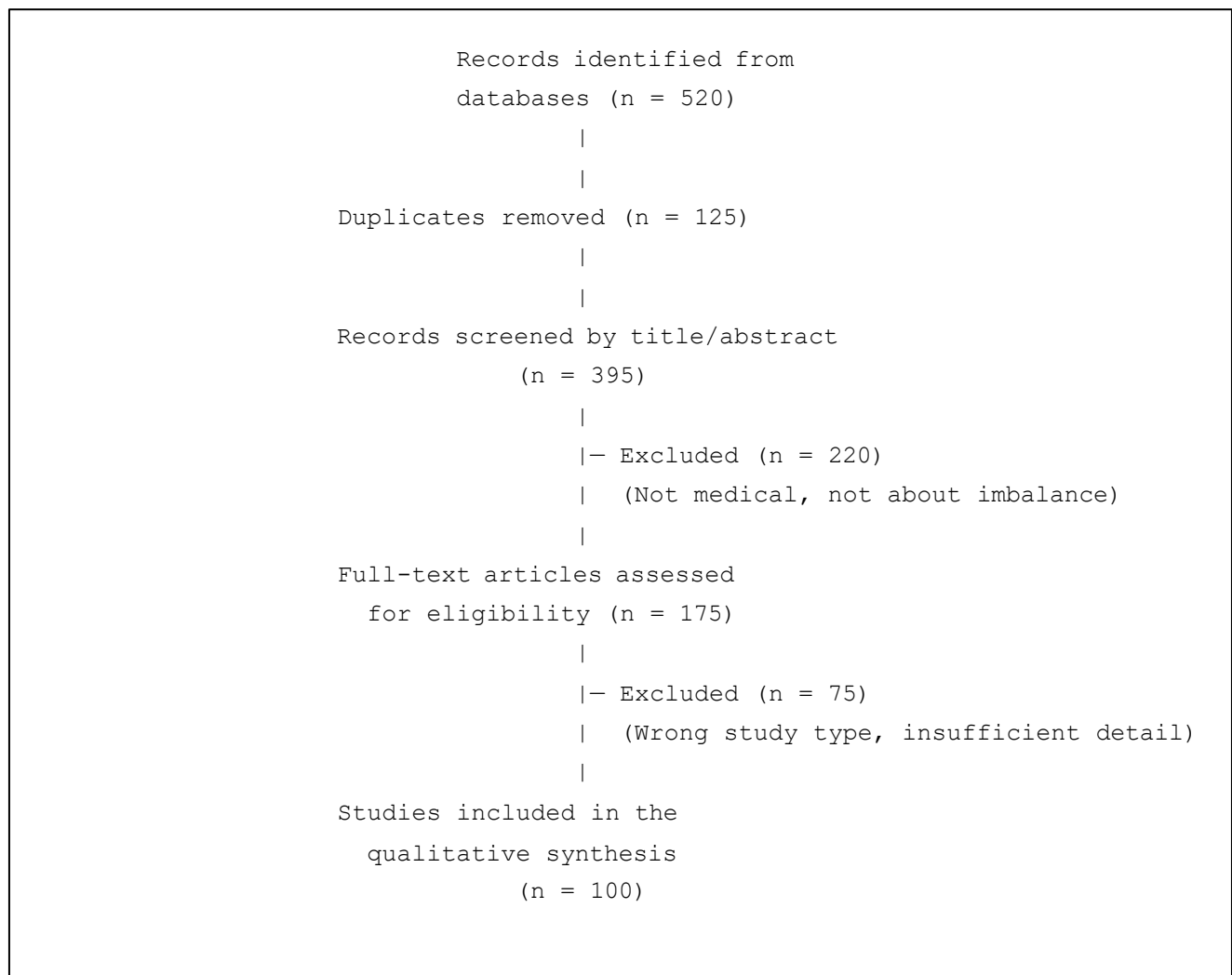
C. Study Selection and Analysis

After removing duplicates, titles and abstracts of the retrieved articles were screened for relevance. The full text of shortlisted papers was then reviewed in detail. Key information was extracted into a structured template, including: medical application area, imbalance ratio, proposed technique, evaluation metrics, and main findings. The studies were thematically grouped to facilitate comparative analysis and synthesis.

D. Organization of the Survey

The reviewed literature is organized to first establish the problem's definition and impact, then dissect its unique challenges in medicine, followed by a detailed examination of solutions, their comparison, and finally, illustrative applications. A critical discussion ties the findings together.

Figure 1: PRISMA-Inspired Study Selection Flow Diagram



3. Imbalanced Data: Definition and Analysis

A. Formal Definition and Characteristics

Imbalanced data refers to a classification scenario where the class distribution is highly skewed. Formally, let C_{maj} represent the set of majority class samples and C_{min} the set of minority class samples. A dataset is considered **severely imbalanced** if:

$$\left| \frac{|C_{\text{min}}|}{|C_{\text{maj}}|} \right| < \delta$$

where δ is a small threshold, often set at 0.2, 0.1, or even lower in medical contexts [5]. For instance, in a cancer screening dataset with 1% prevalence, δ would be approximately 0.01.

The minority class in medical diagnosis often corresponds to rare diseases, early-stage conditions, or specific pathological subtypes. The scarcity of these samples, coupled with their critical importance, defines the core of the problem [6].

B. Impact on Model Performance and Evaluation

Class imbalance induces several critical biases in model development:

- Algorithmic Bias:** Standard classifiers (e.g., Logistic Regression, Decision Trees) trained to minimize overall error rate naturally favor the majority class, as correctly classifying its abundant samples offers the easiest path to lower loss.
- Evaluation Deception:** Metrics like **Accuracy** become meaningless. A model that classifies all mammograms as "healthy" would achieve 99% accuracy if disease prevalence is 1%, while being clinically useless.
- Critical Performance Degradation:** The model's ability to detect the minority class—measured by **Recall (Sensitivity)** or **True Positive Rate**—plummets. This directly translates to an increase in **False Negatives**, the most dangerous error type in diagnostics.

Consequently, the field has adopted a suite of more informative metrics, including **Precision**, **Recall**, **F1-Score**, **Geometric Mean (G-Mean)**, and the **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)** or **Precision-Recall Curve (AUC-PR)**, with the latter being particularly recommended for imbalanced scenarios [7].

4. Challenges of Imbalanced Medical Data

The problem is exacerbated in medicine by domain-specific complexities:

- * **Inherent Low Prevalence:** Many serious diseases are naturally rare in the population.
- * **Cost and Ethical Barriers:** Collecting labeled data for diseases is expensive, time-consuming, and governed by strict ethical protocols.
- * **Data Scarcity and Privacy:** Patient privacy laws (e.g., HIPAA, GDPR) limit data sharing, resulting in small, isolated, and often imbalanced datasets.
- * **Within-Class Heterogeneity and Overlap:** The minority class itself may be diverse (e.g., different cancer subtypes). Furthermore, feature distributions of healthy and sick patients often overlap, especially in early-stage disease, making separation intrinsically difficult even with balanced data [8].
- * **Noisy Labels:** Diagnostic labels in real-world datasets can be imperfect due to human error or ambiguous cases.

These factors collectively make the imbalanced learning problem in medicine particularly acute and resistant to simple solutions.

5. Countermeasures and Proposed Solutions

Solutions can be categorized into two main paradigms: modifying the data distribution or modifying the learning algorithm.

A. Data-Level Solutions (Resampling Techniques)

These methods preprocess the data to create a more balanced distribution before training.

1. **Random Over-Sampling (ROS):** Randomly duplicates minority class instances. While simple, it carries a high risk of **overfitting**, as the model learns from repeated, identical examples [9].

- * **Synthetic Minority Over-sampling Technique (SMOTE):** The seminal algorithm in this category [10]. It creates synthetic samples for the minority class by interpolating between existing instances. It works by:

- * Selecting a minority instance.
- * Finding its k-nearest minority neighbors.
- * Creating a new sample at a random point along the line segment connecting the instance and a randomly chosen neighbor.

* **Limitations:** Can generate noisy, unrealistic samples in regions of class overlap or for high-dimensional data. Variants like **Borderline-SMOTE** (focuses on borderline samples) and **ADASYN** (adaptively generates more samples for harder-to-learn instances) were developed to mitigate these issues [11].

2. **Random Under-Sampling (RUS):** Randomly removes majority class instances. It is computationally efficient but risks **discarding potentially useful information**, which could harm the model's ability to understand the majority class's true structure [9].

B. Algorithm-Level Solutions

These methods adapt the learning algorithm itself to account for the imbalance.

1. **Cost-Sensitive Learning:** This framework assigns a higher misclassification **cost** (or penalty) to errors on the minority class than to errors on the majority class. The learning algorithm is then modified to minimize total cost rather than total error. This is a theoretically sound approach that aligns well with clinical priorities, though it requires careful definition of the cost matrix [12].

2. **Ensemble Learning:** Ensemble methods combine multiple base classifiers to improve generalization. They are naturally more robust to imbalance. Key approaches include:

* **Boosting-based (e.g., AdaBoost, XGBoost):** Sequentially train models where each new model focuses on the instances misclassified by previous ones. This iterative focus naturally gives more weight to the hard-to-classify minority samples.

* **Bagging-based with Resampling (e.g., Balanced Random Forest):** Train each base classifier on a balanced bootstrap sample of the data (e.g., using RUS on the majority class for each tree). This effectively integrates data-level balancing into the ensemble framework [13].

C. Hybrid and Advanced Approaches

State-of-the-art solutions often combine the above strategies. For example, **SMOTEBoost** integrates SMOTE into the AdaBoost algorithm, and **EasyEnsemble** or **BalanceCascade** use sophisticated under-sampling strategies within an ensemble. Recent research also explores deep learning solutions, such as modifying loss functions (e.g., **Focal Loss** [14]) to down-weight the loss assigned to well-classified majority samples, forcing the network to focus on hard minority examples.

D. Comparative Summary

Technique	Category	Key Strengths	Key Limitations	Recommended Context in Medicine
Random Over-Sampling (ROS)	Data-Level	Simple, preserves all info from both classes.	High overfitting risk; does not add new information.	Rarely recommended alone; potentially for tiny, clean minority sets.
SMOTE	Data-Level	Reduces overfitting vs. ROS; adds informative synthetic samples.	Can create noisy/implausible samples in overlapping regions; sensitive to noise.	Medium-sized datasets with clear feature-space structure. Common in medical literature.
Random Under-Sampling (RUS)	Data-Level	Fast; simplifies the problem space.	Discards potentially useful majority data, harming model representation.	Very large datasets where information loss is acceptable.
Cost-Sensitive Learning	Algorithm-Level	Mathematically elegant; aligns with clinical risk (cost).	Requires domain knowledge to set appropriate misclassification costs.	When clinical cost/risk ratios can be reasonably estimated.
Ensemble Methods (e.g., XGBoost, Balanced RF)	Algorithm-Level	Inherently robust; often state-of-the-art performance.	Can be computationally expensive; more hyperparameters to tune.	General-purpose choice for complex, real-world medical datasets.
Hybrid (e.g., SMOTE + Ensemble)	Hybrid	Leverages strengths of both data and algorithm approaches; often most effective.	Increased complexity in implementation and tuning.	For achieving maximum predictive performance and robustness.

6. Applications in Medical Diagnosis

A. Cancer Detection

In breast cancer diagnosis using mammography or histopathology data, severe imbalance is common. Studies consistently show that **SMOTE-enhanced Random Forest or XGBoost classifiers** significantly outperform plain models, achieving superior sensitivity (recall) without substantial loss of specificity [15]. Similar results are reported for lung nodule detection in CT scans and skin lesion classification.

B. Prediction of Chronic Diseases

For conditions like diabetes, heart disease, or sepsis prediction, datasets from electronic health records (EHR) are often imbalanced. Here, **cost-sensitive logistic regression** or **gradient boosting machines (GBM)** have proven effective. The cost matrix can be tuned to reflect the severe consequence of missing a positive case (e.g., a future diabetic complication) versus the cost of a false alert [16].

C. Medical Imaging and Rare Diseases

In diagnosing rare diseases from images (e.g., certain retinal diseases), where positive samples are extremely scarce, advanced techniques like **few-shot learning** and **transfer learning from pre-trained models**, combined with strategic data augmentation (a form of informed over-sampling), are emerging as promising directions [17].

7. Discussion, Conclusion, and Future Directions

A. Discussion

This survey underscores that the "class imbalance problem" in medical diagnosis is, in fact, a **syndrome** comprising data scarcity, algorithmic bias, and inappropriate evaluation. No single solution is universally best. The choice depends on the dataset size, imbalance ratio, feature dimensionality, and available computational resources. The consistent finding across applications is that **conscious handling of imbalance—whether via resampling, algorithmic adjustment, or both—is non-negotiable for developing clinically viable models.**

B. Conclusion

Imbalanced data poses a fundamental threat to the reliability and fairness of ML-based medical diagnostic systems. Relying on standard tools and accuracy metrics leads to models that are dangerously blind to the conditions they are meant to detect. This review has outlined the landscape of solutions, from foundational resampling techniques like SMOTE to advanced algorithmic adaptations like cost-sensitive ensemble learning. The evidence strongly suggests that **hybrid approaches**, which combine the complementary strengths of data manipulation and robust learning algorithms, provide the most effective pathway to building accurate, sensitive, and trustworthy diagnostic aids.

C. Future Directions

Several open challenges and research frontiers remain:

1. **Beyond Binary Imbalance:** Addressing **multi-class imbalance**, where several rare conditions must be distinguished from a large "normal" class and from each other.
2. **Integration with Complex Data:** Developing methods tailored for highly imbalanced **multimodal data** (e.g., combining EHR, images, and genomics).
3. **Explainability and Trust:** Ensuring that models built on resampled or cost-adjusted data remain interpretable to clinicians.
4. **Dynamic and Cost-Aware Learning:** Creating frameworks that can adaptively learn cost matrices from clinical outcomes or operational constraints.
5. **Benchmarking and Standards:** The community would benefit from standardized, publicly available, severely imbalanced medical benchmarks to fairly compare new algorithms.

Addressing these challenges will be crucial for translating the promise of medical AI into safe, effective, and equitable clinical practice.

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